

Problem 4.17

Smith has just filed his income tax return and is expecting to receive, in 60 days, a refund check of 1000.

(a) The tax service that helped him fill out his return offers to buy Smith's refund check for 850. What annual simple interest rate is implied?

(b) Smith negotiates and sells his refund check for 900. What annual simple interest rate does this correspond to?

(c) Smith deposits the 900 in an account which earns simple interest at annual rate 9%. How many days would it take from the time of his initial deposit of 900 for the account to reach 1000?

Solution



For simple interest we use

$$A(t) = P(1 + it),$$

where t is measured in years.

For 60 days, we take

$$t = \frac{60}{365}.$$

(a) If Smith sells the refund check for 850, then

$$1000 = 850 \left(1 + i \frac{60}{365} \right).$$

Solving for i ,

$$i = \left(\frac{1000}{850} - 1 \right) \frac{365}{60} = 1.07353.$$

Therefore, the annual simple interest rate is

$$\boxed{107.35\%}.$$

(b) If Smith sells the refund check for 900, then

$$1000 = 900 \left(1 + i \frac{60}{365} \right).$$

So,

$$i = \left(\frac{1000}{900} - 1 \right) \frac{365}{60} = 0.67592.$$

Therefore, the corresponding annual simple interest rate is

$$\boxed{67.592\%}.$$

(c) Now Smith deposits 900 into an account earning 9% simple interest. We want the time t in days for the account to reach 1000. Then

$$1000 = 900 \left(1 + 0.09 \frac{t}{365} \right).$$

Solving for t ,

$$\frac{1000}{900} - 1 = 0.09 \frac{t}{365},$$

so

$$t = \left(\frac{1000}{900} - 1 \right) \frac{365}{0.09} = 450.6173.$$

Thus it would take approximately

$$\boxed{451 \text{ days}}.$$